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0775/1/2025
F. MATHS A/L

SOUTH WEST REGIONAL MOCK EXAMINATION

GENERAL EDUCATION

The Teachers' Resource Unit (TRU) in collaboration with the Regional Pedagogic Inspectorate of Science and the South West Association of Mathematics Teachers (SWAMT)	Subject Code 0755	Paper Number 1
CANDIDATE NAME ..... CANDIDATE NUMBER ..... CENTRE NUMBER	Subject Title FURTHER MATHEMATICS	
ADVANCED LEVEL	DATE: 26/3/2025 DAY: Wednesday Morning	

Time Allowed: One hour thirty minutes

INSTRUCTIONS TO CANDIDATES:

1. USE A SOFT HB PENCIL THROUGH OUT THIS EXAMINATION.
2. DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Before the Examination begins:

3. Check that this question booklet is headed "Advanced Level – 0775 Further Mathematics, Paper 1".
4. Insert the information required in the spaces provided above.
5. Without opening the booklet, pull out the answer sheet carefully from inside the front cover of this booklet. Take care that you do not crease or fold the answer sheet or make any marks on it other than those asked for in these instructions.
6. Insert the information required in the spaces provided on the answer sheet using your HB pencil:

Candidate Name, Centre Number, Candidate Number, Subject Code Number and Paper Number.

How to answer questions in this Examination:

7. Answer ALL the 50 questions in this examination. All questions carry equal marks.
8. Non-programmable calculators are allowed.
9. For each question there are four suggested answers, A, B, C, and D. Decide which answer is correct. Find the number of the question on the Answer sheet and draw a horizontal line across the letter to join the square brackets for the answer you have chosen. For example, if C is your correct answer, mark C as shown below:

[A] [B] [C] [D]

10. Mark only one answer for each question. If you mark more than one answer, you will score zero for that question. If you change your mind about an answer, erase the first mark carefully, and then mark your new answer.
11. Avoid spending much time on any question. If you find a question difficult, move to the next question. You can come back to this question later.
12. Do all rough work in this booklet using, where necessary, the blank spaces in the question booklet.
13. Mobile phones are **NOT ALLOWED** in the examination room.
14. You must not take this booklet and answer sheet out of the examination room. All question booklets and answer sheets will be collected at the end of the examination.

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1. $\frac{7x^3}{(x^2+5)^2}$ decomposed in partial fractions, where a, b, c and d are real constants is

A	$\frac{ax+b}{x^2+5}+\frac{cx+d}{(x^2+5)^2}$
B	$\frac{ax+b}{x^2+5}+\frac{cx^2+d}{(x^2+5)^2}$
C	$\frac{ax+b}{x^2+5}+\frac{cx+d}{x^2+5}$
D	$\frac{ax^2+b}{x^2+5}+\frac{cx+d}{(x^2+5)^2}$

2. Two vectors $\vec{a}$ and $\vec{b}$ are given as $\vec{a}=\vec{i}+2\vec{j}+\vec{k}$ and $\vec{b}=3\vec{i}+\vec{j}+2\vec{k}$. The vector product $\vec{a}\times\vec{b}=$

A	$3\vec{i}-\vec{j}-5\vec{k}$
B	$3\vec{i}+\vec{j}-5\vec{k}$
C	$3\vec{i}+\vec{j}+5\vec{k}$
D	$3\vec{i}-5\vec{j}-5\vec{k}$

3. Which of the following defines an equation of a curve that does NOT have a vertical asymptote?

A	$y=\frac{x^3+2x^2}{x+2}$
B	$y=\frac{x^2-x-4}{x+2}$
C	$y=\frac{x^3+x}{x+2}$
D	$y=\frac{x^3-x}{x+2}$

4. The general solution of the differential equation $\sin x\frac{dy}{dx}+y\cos x=\cos x$ is:

A	$y=-1+k\,\text{cosec}\,x$
B	$y=1+k\,\text{cosec}\,x$
C	$y=(-1+k)\,\text{cosec}\,x$
D	$y=(1+k)\,\text{cosec}\,x$

5. Given that $f(x)=\begin{cases}x^2, \text{for } x<1\\ 2x-1, \text{for } 1\le x\le 3,\\ 7-x, \text{for } x>3\end{cases}$ a value of x for which $f(x)$ is discontinuous is

A	B	C	D
0	1	3	2

6. Given the statements p and q, $\sim(p\Rightarrow\sim q)\equiv$

A	$\sim p\wedge q$
B	$p\wedge\sim q$
C	$p\wedge q$
D	$\sim p\wedge\sim q$

7. Given the non-singular square matrices N and M, and non-zero column vectors x and y such that $(NM^{-1})x=y$, then $x=$

A	$MN^{-1}y$
B	$N^{-1}My$
C	$M^{-1}Ny$
D	$NM^{-1}y$

8. $\int_0^{\ln 2} e^{\sinh x}\cosh x\,dx=$

A	$e^{\frac{3}{4}}-1$
B	$e^{\frac{5}{4}}-e$
C	$e^{\frac{3}{2}}-1$
D	$e^{\frac{1}{4}}-1$

9. The general solution of the differential equation $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+3y=0$, where P and Q are real constants is:

A	$e^{3x}(P\cos x+Q\sin x)$
B	$e^{x}(P\cos 3x+Q\sin 3x)$
C	$Pe^{x}+Qe^{3x}$
D	$(P+Qx)e^{-4x}$

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10. The polar form of the Cartesian equation $(x^2+y^2)^2 = a^2xy$ is

A	$r^2 = \frac{a^2}{2}\sin 2\theta$
B	$r^2 = a^2 \sin 2\theta$
C	$r^2 = \frac{a^2}{2}\sin\theta$
D	$r^2 = \frac{a^2}{4}\sin 2\theta$

11. The function below that is continuous on the interval $]0,\infty[$ is

A	$f(x) = \frac{x^2+3}{\cos x}$
B	$g(x) = \frac{\sqrt{x^2+1}}{x-2}$
C	$h(x) = \frac{x^3+x-1}{x+2}$
D	$k(x) = \frac{\lvert\sin x\rvert}{x-1}$

12. The function $f(x) = \sqrt{\frac{x-1}{x+1}}$ is continuous on the interval

A	$]-\infty,1]$
B	$[-1,1]$
C	$]-1,1[$
D	$]-\infty,1[$

13. If p and q are mathematical statements, then the statement $\sim q \Rightarrow p \equiv$

A	$\sim p \Rightarrow q$
B	$q \Rightarrow \sim p$
C	$\sim q \Rightarrow \sim p$
D	$q \Rightarrow p$

14. If $|a| < \prod$, then $\int_{-a}^{a} \cos|x|dx =$

A	$2\int_{-a}^{a} \cos x dx$
B	0
C	$\int_{0}^{a} \cos x dx$
D	$2\int_{0}^{a} \cos x dx$

15. A force $F = (2\vec{i}-3\vec{j}+\vec{k})N$ acts through the point with position vector $(\vec{i}-3\vec{j}-2\vec{k})m$. Its moment about the point with position vector $-\vec{i}+\vec{j}+3\vec{k}$ is

A	$-19\vec{i}-12\vec{j}+2\vec{k}$
B	$19\vec{i}-12\vec{j}+2\vec{k}$
C	$-19\vec{i}+12\vec{j}+2\vec{k}$
D	$19\vec{i}-12\vec{j}-2\vec{k}$

16. Given that $|Z| = 5$, then the greatest value of $|Z-2|$ is

A	B	C	D
2	3	5	7

17. The group of complex numbers under multiplication $(C,\times)$, where $C = \{1, i, -1, -i\}$ is a cyclic group. The generator of this group is

A	B	C	D
1	i	-1	$-i$

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18. Which of the following congruence equations has a solution?

A	$24x \equiv 18(\text{mod } 32)$
B	$5x \equiv 4(\text{mod } 10)$
C	$15x \equiv 6(\text{mod } 25)$
D	$16x \equiv 12(\text{mod } 36)$

19. The equation $25x^2 - y^2 - 100x - 2y + 76 = 0$ Represents

A	a parabola
B	a circle
C	a hyperbola
D	an ellipse

20. The focus of the parabola $x^2 = 16(y+1)$ is at the point

A	B	C	D
(0,3)	(0,4)	(3,0)	(4,0)

21. Two smooth spheres each of mass m collide when moving with velocities $\left(\vec{i}+3\vec{j}\right)ms^{-1}$ and $\left(-2\vec{i}+\vec{j}\right)ms^{-1}$ respectively. If the line joining their centres is parallel to $\vec{i}$ and the coefficient of restitution between the spheres is $\frac{1}{3}$, then the velocities of the spheres after impact is respectively

A	$\left(\vec{i}-\vec{j}\right)ms^{-1}$ and $-\vec{j}ms^{-1}$
B	$\left(-\vec{i}-3\vec{j}\right)ms^{-1}$ and $-\vec{j}ms^{-1}$
C	$\left(\vec{i}+3\vec{j}\right)ms^{-1}$ and $\vec{j}ms^{-1}$
D	$\left(-\vec{i}+3\vec{j}\right)ms^{-1}$ and $\vec{j}ms^{-1}$

22. Given that kx^2e^{2x} is the particular integral for the differential equation $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = e^{2x}$ where k is a constant, then the value of k is

A	B	C	D
$\frac{1}{2}$	2	$-\frac{1}{2}$	3

23. A recursive sequence $(x)_n, n \geq 1$ is defined by $x_{n+1} = (-x_n + 4)$ and $x_1 = 2$. The sequence is

A	decreasing
B	constant
C	increasing
D	alternating

24. The matrix representing the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (2x - y + z, x + y, x - 2y + z)$ is

A $\begin{bmatrix} 2 & 1 & 1 \\ -1 & 1 & -2 \\ 1 & 0 & 1 \end{bmatrix}$

B $\begin{bmatrix} 1 & -1 & 2 \\ 1 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$

C $\begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$

D $\begin{bmatrix} 1 & -2 & 1 \\ 1 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}$

25. The polar equation of the hyperbola $xy = 16$ is

A	$r^2 = 16\cos ec\theta$
B	$r^2 = 16\cos ec2\theta$
C	$r^2 = 8\cos ec2\theta$
D	$r^2 = 32\cos ec2\theta$

26. A group G has subgroups $\{a, d\}$, $\{b, c, d, f\}$, $\{c, d, f\}$. The identity element of G is

A	B	C	D
c	f	d	a

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27. A particle P moves with constant angular velocity ω on the polar equation $r = ae^{\frac{\theta}{k}}$ where a and k are positive constants. The radial component of the velocity of the particle P is

A	B	C	D
$\frac{a\omega r}{k}$	ωr	$\frac{\omega r}{k}$	$k\omega r$

28. Given that the energy equation of a compound pendulum is $\frac{a}{4}\left(\frac{d\theta}{dt}\right)^2 = g\cos\theta$, where a and g are constants, its period is

A	$\frac{\Pi}{2}\sqrt{\frac{a}{g}}$
B	$\Pi\sqrt{\frac{a}{g}}$
C	$2\Pi\sqrt{\frac{a}{g}}$
D	$\Pi\sqrt{\frac{a}{2g}}$

29. The series $\sum_{r=1}^{\infty}\frac{1}{4^r}$ is

A	Divergent
B	Convergent
C	Telescopic
D	Oscillating

30. A particle moving with velocity $a\vec{i}\ ms^{-1}$ receives an impulse which changes its velocity to $\left(\vec{i}+\vec{k}\right)ms^{-1}$. If the particle is deflected through $\frac{\Pi}{4}$, the value of a is

A	B	C	D
$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	1	$\sqrt{2}$

31. A particle P describes simple harmonic motion with O as the centre and has a speed of $6\ m\ s^{-1}$ at a distance of $1m$ from O and a speed of $2\ m\ s^{-1}$ at a distance of 3 m from O. Its amplitude is

A	B	C	D
$2\sqrt{2}$m	$\sqrt{10}m$	$10m$	$\sqrt{5}m$

32. The series $\sum_{r=1}^{\infty}\frac{1}{r^n}$ converges when

A	B	C	D
$n>0$	$n<1$	$n=1$	$n>1$

33. Given that $7x \equiv 1(\text{mod}\,9)$, then $x =$

A	B	C	D
7	6	5	4

34. $\int\frac{1}{\sqrt{x^2+4}}dx =$

A	$\cosh^{-1}\left(\frac{x}{4}\right)+c$
B	$\sinh^{-1}\left(\frac{x}{4}\right)+c$
C	$\cosh^{-1}\left(\frac{x}{2}\right)+c$
D	$\sinh^{-1}\left(\frac{x}{2}\right)+c$

35. A random variable X has a Poisson distribution with parameter 4. $P(X<2) =$

A	$4e^{-4}$
B	$1-4e^{-4}$
C	$1+4e^{-4}$
D	$5e^{-4}$

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36. Given that $y = \cos ech4x$, then $\frac{dy}{dx} =$

A	$-4\cos ech4x\coth 4x$
B	$4\cos ech4x\coth 4x$
C	$-\cos ech4x\coth 4x$
D	$\cos ech4x\coth 4x$

37. The number $\frac{2}{7}$ can be expressed as a decimal as $0.285714285714285714\ldots$. The 307th digit is

A	B	C	D
8	7	4	2

38. The polar coordinates of a particle P at time t are (r,θ) where $r = 3t^2$ and $\theta = t^2$. The radial component of the acceleration of P at time $t = 1$ second is

A	B	C	D
18	15	12	6

39. If $\tanh y = \frac{x}{2}$, then $y =$

A	$\frac{1}{2}\ln\left(\frac{2-x}{2+x}\right)$
B	$\frac{1}{2}\ln\left(\frac{2+x}{2-x}\right)$
C	$\frac{1}{2}\ln\left(\frac{1-2x}{1+2x}\right)$
D	$\frac{1}{2}\ln\left(\frac{1+2x}{1-2x}\right)$

40. The mean value of $f(x) = e^x$ in the interval $1 \le x \le 5$ is

A	$e^5 - e$
B	$\frac{1}{4}(e^5 - e)$
C	$\frac{1}{5}(e^5 - e)$
D	$\frac{1}{6}(e^5 - e)$

41. The centre of symmetry of the curve $y = \frac{1}{x+3}$ is at the point

A	B	C	D
(3,0)	(−3,0)	(0,0)	(0,3)

42. When x is large and positive, $\frac{1}{\text{sech}\, x} \approx$

A	$\frac{1}{2}\left(e^x + e^{-x}\right)$
B	e^x
C	$\frac{1}{2}e^{-x}$
D	$\frac{1}{2}e^x$

43. $\frac{1}{e^{-2i\theta}} =$

A	$\frac{1}{2}(\cos 2\theta - i\sin 2\theta)$
B	$\frac{1}{2}(\cos 2\theta + i\sin 2\theta)$
C	$\cos 2\theta - i\sin 2\theta$
D	$\cos 2\theta + i\sin 2\theta$

44. The distance between the planes $3x - y + 2z = -5$ and $3x - y + 2z = -15$ is

A	$\frac{15\sqrt{14}}{7}$
B	$\frac{15\sqrt{14}}{14}$
C	$\frac{10\sqrt{14}}{7}$
D	$\frac{5\sqrt{14}}{7}$

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45. A continuous random variable Y is uniformly distributed in the interval $25 \leq y \leq 47$. The median of this variable is

A	B	C	D
47	36	22	11

46. The moment of inertia of a rigid body of mass $4m$ about a given axis is $18ma^2$. The radius of gyration about this axis is

A	$3a^2\sqrt{2}$
B	$3a\sqrt{2}$
C	$\frac{3a\sqrt{2}}{2}$
D	$\frac{3a^2\sqrt{2}}{2}$

47. The position vector of a particle P of mass $3kg$ is $\left(\vec{j}+2\vec{k}\right)m$ and its velocity is $\left(\vec{i}-\vec{j}\right)ms^{-1}$. Its momentum is

A	$-6\vec{i}+6\vec{j}-3\vec{k}$
B	$-2\vec{i}+2\vec{j}-3\vec{k}$
C	$2\vec{i}+2\vec{j}-3\vec{k}$
D	$6\vec{i}+6\vec{j}-3\vec{k}$

48. The vector equation of a curve is $\vec{r}=2t\vec{i}+3t^2\vec{j}$. The Cartesian equation of the curve is

A	$3y^2=4x$
B	$3y=4x^2$
C	$4y=3x^2$
D	$4y^2=3x$

49. A particle P describes simple harmonic motion of amplitude $2m$. In performing one complete oscillation, its displacement is

A	B	C	D
$10\,m$	$6\,m$	$5\,m$	$2\,m$

50. The normal approximation to the Poisson distribution of the random variable X with variance $\frac{1}{2}$ is

A	$X \sim N\left(\frac{1}{2},\frac{1}{4}\right)$
B	$X \sim N\left(\frac{1}{2},\frac{1}{2}\right)$
C	$X \sim N\left(\frac{1}{4},\frac{1}{8}\right)$
D	$X \sim N\left(\frac{1}{4},\frac{1}{4}\right)$

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